Block 3: Measuring Topology via Quantum Circuits

Seminar notes

August 4, 2026

1 Introduction

In Block 2, we mapped the fermionic Kitaev chain to a system of qubits using the Jordan-Wigner transform. The resulting Hamiltonian is a 1D transverse-field XY model:

$$H = -\frac{\mu}{2}\sum_{j=0}^{L-1} Z_j + \frac{t-\Delta}{2}\sum_{j=0}^{L-2} X_j X_{j+1} + \frac{t+\Delta}{2}\sum_{j=0}^{L-2} Y_j Y_{j+1}. \tag{1}$$

For Block 3, our goal is to simulate this system using a quantum circuit framework (Qiskit). This requires two main steps:

1. **State Preparation:** Initializing the qubits into the many-body ground state $|\Phi_0^{(+)}\rangle$.
2. **Measurement:** Evaluating local and non-local (string) observables to detect the topological phase.

2 Approaches to Ground State Preparation

Preparing the exact ground state of a strongly correlated many-body system is generally hard, but the Kitaev chain is integrable (it maps to free fermions). We have a few distinct paths to prepare the state on a quantum computer.

2.1 Approach 1: Exact Free-Fermion State Preparation (Algebraic)

Since the system is non-interacting, its ground state is a Slater determinant. Any Slater determinant can be prepared exactly using a polynomial-depth quantum circuit.

- **Method:** We diagonalize the real-space BdG matrix to find the Bogoliubov transformation matrix W. Using QR or SVD-like decompositions, W can be factored into a sequence of nearest-neighbor Givens rotations.
- **Circuit:** These rotations map directly to two-qubit fermionic simulation gates (e.g., $XX+YY$ rotations, known as fSim gates).
- **Pros:** Prepares the state exactly with 100% theoretical fidelity. Provides a perfect theoretical baseline.
- **Cons:** The circuit depth scales linearly with L (or $L^2/2$ for full basis transforms), which can be deep for NISQ devices.

2.2 Approach 2: Variational Quantum Eigensolver (VQE)

VQE is a hybrid quantum-classical heuristic. We define a parameterized quantum circuit $U(\vec{\theta})$ (the ansatz), prepare the state $|\psi(\vec{\theta})\rangle = U(\vec{\theta})|0\rangle$, measure the energy $E(\vec{\theta}) = \langle\psi|H|\psi\rangle$, and use a classical optimizer to minimize E.

- **Method:** We can use a Hamiltonian Variational Ansatz (HVA), which applies alternating layers of $e^{-i\theta_1 Z_j}$, $e^{-i\theta_2 X_j X_{j+1}}$, and $e^{-i\theta_3 Y_j Y_{j+1}}$ gates.
- **Pros:** The circuits are typically very shallow and resilient to certain systematic errors, making this the standard approach for near-term (NISQ) hardware.
- **Cons:** It is an approximation. Finding the global minimum can be difficult (barren plateaus), and the resulting state has a fidelity strictly less than 1.

2.3 Approach 3: Exact Diagonalization Vector Loading (StateVector)

For small systems ($L \leq 12$), we can simply diagonalize the $2^L \times 2^L$ qubit Hamiltonian classically to find the exact state vector, and then forcefully load this vector into the Qiskit simulator.

- **Pros:** Trivial to implement using `qiskit.QuantumCircuit.initialize()`.
- **Cons:** Not a true quantum algorithm. The initialization gate compiles to an exponentially deep circuit, so it is useless for real hardware. It only serves as a classical simulation tool to verify observables.

Selected Path (Revised for Week 5): Based on the Week 5 Execution Plan, we must strictly deprecate **Approach 3 (StateVector)** and exact diagonalization using dense matrices. The architecture must fully transition to gate-based quantum circuit simulation. We will implement **Approach 2 (VQE)** with a parameterized quantum circuit (PQC) using RY and CX/CZ gates. The state output must achieve a fidelity of $F > 0.99$ compared to the exact state for small systems ($L \leq 8$) before proceeding to measurements.

3 Measuring Observables

Once the ground state $|\Phi\rangle$ is prepared, we must measure the signatures of topology.

3.1 Edge Observables (Local)

In the topological phase, Majorana zero modes are localized at the edges. At the sweet spot ($t = \Delta, \mu = 0$), the Hamiltonian reduces to $H = t \sum Y_j Y_{j+1}$, completely decoupling the boundary operators X_0 and X_{L-1}. Therefore, the edge Majorana modes are directly related to the local Pauli observables:

$$\mathcal{O}_{\text{edge},L} = X_0, \qquad \mathcal{O}_{\text{edge},R} = X_{L-1}. \tag{2}$$

When evaluating $\langle\Phi_0|X_0|\Phi_0\rangle$, however, we must be careful: the absolute ground state has a definite parity, and a single X operator flips parity. To observe the mode, we instead look at correlations or state transfers.

3.2 String Order Parameters (Non-Local)

The defining feature of a 1D topological phase is that it cannot be characterized by a local order parameter (like magnetization). Instead, it possesses hidden long-range order, detectable via a **String Order Parameter (SOP)**. For the Kitaev chain mapped to qubits, the topological phase corresponds to the ferromagnetic phase of the transverse-field Ising/XY model. The appropriate string order parameter spanning sites i and j is:

$$\mathrm{SOP}(|i-j|) = \langle X_i \left(\prod_{k=i+1}^{j-1} Z_k \right) X_j \rangle. \tag{3}$$

In the trivial phase ($|\mu| > 2t$), this expectation value decays exponentially to 0. In the topological phase ($|\mu| < 2t$), this string correlation attains a non-zero long-range asymptotic value as $|i-j| \to \infty$.

3.3 Shot-Based Sampling Requirement

Analytical expectation values from statevectors must be discarded. The measurement protocol must execute shot-based sampling (e.g., `shots=8192`).

- **Local $\langle X_0 \rangle$:** Apply an H gate on qubit 0 to rotate to the X-basis, then measure in the Z-basis. This local parity-odd expectation should vanish in a parity eigenstate.
- **Edge string $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$:** Apply H gates to qubits 0 and $L-1$, measure every qubit in the Z basis, then multiply the measured Z eigenvalues in post-processing. For $L = 4$ this is $\langle X_0 Z_1 Z_2 X_3 \rangle$.

Post-processing code will calculate the joint parity from the 0/1 counts, extract the statistical expectation values, and compute the standard error.

4 Implementation in Qiskit (block3.py)

Following the theoretical framework and the Week 5 plan, the Python implementation (`block3.py`) explicitly deprecates exact diagonalization as a solution. Instead, it relies on gate-based quantum simulation utilizing the `qiskit_aer` module.

4.1 VQE Ansatz Construction

We utilize Qiskit's `EfficientSU2` library to create a hardware-efficient ansatz. The circuit consists of alternating layers of single-qubit $RY(\theta)$ rotations and two-qubit linear entanglement gates (CX).

4.2 Ground State Fidelity Validation

A multi-start optimization loop using the COBYLA algorithm minimizes the energy expectation value. To ensure the heuristic VQE result is scientifically valid before proceeding to measurements, a strict checkpoint is enforced: the fidelity $F = |\langle \psi_{ED} | \psi_{VQE} \rangle|^2$ must exceed 0.99. Crucially, in the topological phase ($\mu = 0$), the ground state and the first excited state are near-degenerate. The code correctly handles this by summing the overlap probabilities across the degenerate subspace to accurately evaluate the achieved fidelity.

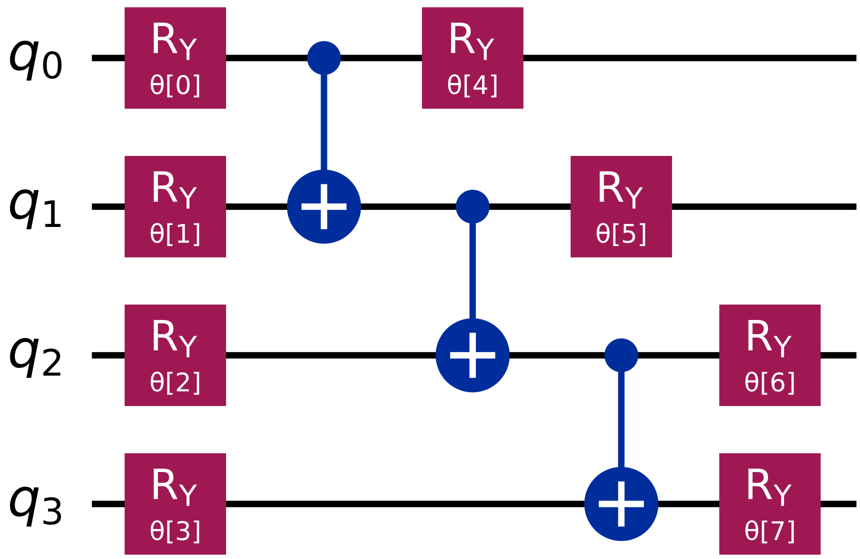

Figure 1: The raw VQE ansatz circuit diagram for $L = 4$ (reps=1 shown for clarity).

4.3 Optimizer Choice

The ground-state preparation is split between two SciPy optimizers depending on context. The Week 5 convergence study uses the gradient-based `L-BFGS-B`, whereas the multi-start preparation and the parity-constrained μ-sweep use the derivative-free `COBYLA`. The reasoning:

- **No analytic gradient.** We supply no gradient, so `L-BFGS-B` would fall back on finite differences, costing $O(n_\theta)$ extra circuit evaluations per step (here $n_\theta = (\text{reps} + 1)\, L = 16$). `COBYLA` is derivative-free: it builds a linear trust-region model from a simplex of points and needs no gradient passes.

- **Noise robustness.** This framework targets shot-based and, ultimately, NISQ-noisy estimators. Finite-difference gradients divide stochastic noise by the step size, which destabilizes `L-BFGS-B`'s line search and curvature updates; `COBYLA`'s trust-region model degrades far more gracefully. This is the standard motivation for derivative-free (or `SPSA`) optimizers in VQE.

- **Caveat.** For the *noiseless statevector* cost actually used in the sweep, the objective is smooth and `L-BFGS-B` would converge faster and more accurately; `COBYLA` is retained for forward-compatibility with the noisy Block 4 estimators. Empirically it reaches fidelity ≈ 1 with zero random restarts across all 81 sweep points.

Alternatives considered: `SPSA` (the standard for noisy hardware VQE, two evaluations per iteration regardless of n_θ), and parameter-shift gradients feeding `L-BFGS-B`/Adam (exact circuit gradients at $2n_\theta$ evaluations per step). Both are unnecessary for the $L = 4$ noiseless problem.

4.4 Measurement Circuits

Expectation values are derived via classical post-processing of 8192 statistical bitstring samples. For the local Majorana observable $\langle X_0 \rangle$, an H gate is applied to qubit 0 immediately before measurement to switch the basis from Z to X.

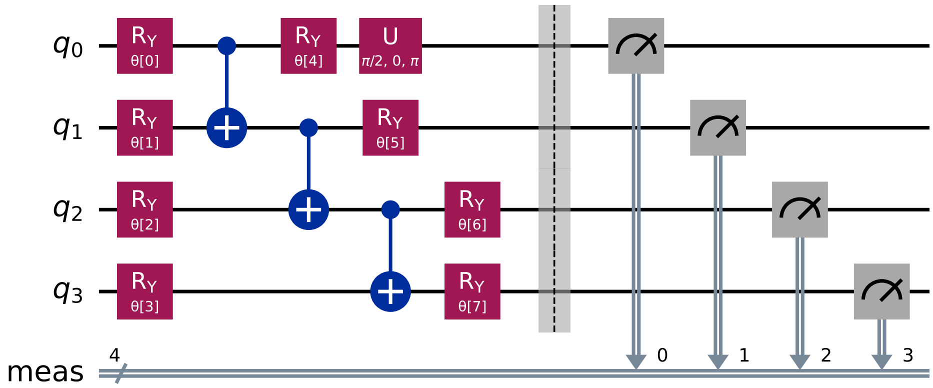

Figure 2: The measurement circuit for the local observable $\langle X_0 \rangle$.

For the non-local edge string $\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle$, the two endpoint qubits are rotated with Hadamard gates and all qubits are measured in the Z basis. The product of the measured signs gives one shot sample of the full string.

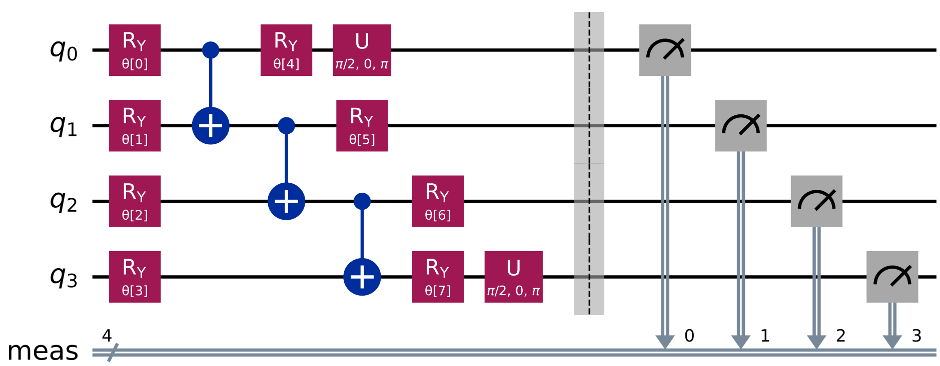

Figure 3: The measurement circuit for the parity-preserving edge string $\langle X_0 Z_1 Z_2 X_3 \rangle$.

4.5 Results

Testing this framework at $L = 4$ yields successful state preparation with fidelity ≥ 0.998 across the three representative phases: Topological ($\mu = 0$), Critical ($\mu = 2t$), and Trivial ($\mu = 3t$). The extracted observables confirm the expected behaviors:

As shown above, the non-local edge-string signal is large in the topological phase and strongly suppressed in the trivial phase, serving as a clear, detectable signature of the Kitaev chain's underlying topology on a quantum processor.

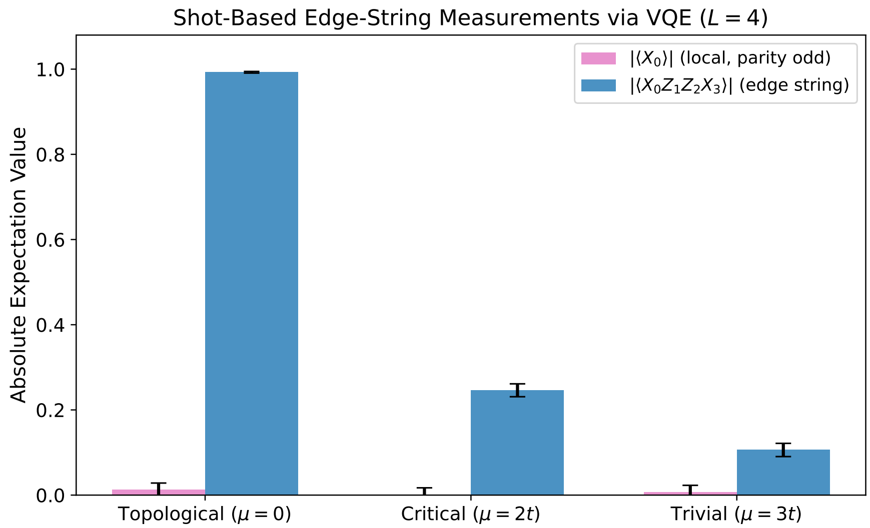

Figure 4: Shot-based VQE expectation values at representative phase points.

5 Week 5 Presentation Audit and Reconciliation

This section records the comparison between the Week 5 presentation (`presentation/week5/slides.tex`), the current note, the scripted implementation (`src/block3.py`), and the notebook used to generate the Week 5 presentation figures (`src/block3_week5.ipynb`). The key point is that the Week 5 presentation is not a direct rendering of the broader `block3.py` workflow. It is a narrower, sweet-spot demonstration focused on gate-based measurement of a specific topological correlator.

5.1 What the Week 5 Presentation Actually Shows

The Week 5 deck is organized around four claims:

1. Move from exact matrix math to gate-based simulation.
2. Prepare a ground state with a hardware-efficient VQE ansatz.
3. Identify the correct non-local string observable for the isolated edge Majoranas.
4. Measure that observable by basis rotations and computational-basis sampling.

The presentation specializes to the topological sweet spot

$$t = \Delta, \qquad \mu = 0, \tag{4}$$

where the Jordan-Wigner Hamiltonian reduces to

$$H = \sum_{j=0}^{L-2} Y_j Y_{j+1}. \tag{5}$$

For $L = 4$, the deck interprets this as a pure nearest-neighbor YY coupling chain. In the Majorana picture used by the slides, these bonds leave two boundary Majoranas isolated, labelled there as $\gamma_{0,2}$ and $\gamma_{3,1}$. The central observable is therefore the boundary Majorana correlator

$$i\gamma_{0,2}\gamma_{3,1} = i(Y_0)(Z_0 Z_1 Z_2 X_3) = -X_0 Z_1 Z_2 X_3. \tag{6}$$

Up to the overall minus sign, the measured Pauli string is therefore

$$O_{\rm corr} = X_0 Z_1 Z_2 X_3. \tag{7}$$

The measurement circuit applies Hadamards to qubits 0 and 3 so that the two X factors are converted to Z-basis measurements:

$$HXH = Z. \tag{8}$$

The remaining $Z_1 Z_2$ factors are already directly measurable in the computational basis. The reported numerical evidence is that local edge expectations such as $\langle Y_0 \rangle$ vanish, while the non-local correlator has magnitude close to 1.

5.2 What the Notebook Implements

The notebook `src/block3_week5.ipynb` is the closest source to the Week 5 figures. It does not implement the full (μ, t, Δ) Hamiltonian used in `block3.py`. Instead, it builds the pure-YY Hamiltonian directly:

$$H_{\rm notebook} = \sum_{j=0}^{L-2} Y_j Y_{j+1}, \qquad L = 4. \tag{9}$$

The ansatz is a real hardware-efficient circuit:

$$U(\theta) = \texttt{efficient\_su2}(L,\ \texttt{su2\_gates=['ry']}, \texttt{entanglement='linear'},\ \texttt{reps} = 3). \tag{10}$$

The optimizer is L-BFGS-B, not COBYLA. The cost function is also not just the physical energy. It adds a parity-selection term:

$$C(\theta) = \langle H \rangle_\theta - \lambda \langle P \rangle_\theta, \qquad P = Z_0 Z_1 Z_2 Z_3, \qquad \lambda = 0.1. \tag{11}$$

This explicitly biases the VQE result toward the even-parity sector. This is important because the topological sweet spot has a degenerate or nearly degenerate ground-state subspace. The notebook then checks the VQE state against the exact degenerate ground subspace:

$$F_{\rm subspace} = \sum_{n:\ E_n \approx E_0} |\langle E_n | \psi_{\rm VQE} \rangle|^2. \tag{12}$$

This fidelity check is still exact diagonalization, but it is used as a small-system validation benchmark, not as the preparation method.

The notebook measurement cell samples three observables with `AerSimulator` and `shots=8192`:

$$\langle Y_0 \rangle, \tag{13}$$

$$\left\langle X_{L-1} \prod_{j=0}^{L-2} Z_j \right\rangle, \tag{14}$$

$$\left\langle X_0 \left(\prod_{j=1}^{L-2} Z_j \right) X_{L-1} \right\rangle. \tag{15}$$

For $L = 4$, the final observable is exactly

$$\langle X_0 Z_1 Z_2 X_3 \rangle. \tag{16}$$

This final correlator is the actual topological signature shown in the presentation.

5.3 Comparison with `block3.py`

The file `src/block3.py` is more general in intent, but it is not the same as the Week 5 notebook or slides.

- **Hamiltonian:** `block3.py` constructs the full Jordan-Wigner Hamiltonian

$$H = -\frac{\mu}{2}\sum_j Z_j + \frac{t-\Delta}{2}\sum_j X_j X_{j+1} + \frac{t+\Delta}{2}\sum_j Y_j Y_{j+1}.$$

  The notebook fixes the sweet spot and directly constructs only $\sum_j Y_j Y_{j+1}$.

- **Optimizer:** `block3.py` uses multi-start COBYLA. The notebook uses a single L-BFGS-B run with a larger iteration budget.

- **Parity handling:** `block3.py` handles degeneracy only when evaluating fidelity, by summing overlap with the first two eigenstates if they are nearly degenerate. The notebook adds the parity penalty directly to the VQE cost function, so parity selection is part of optimization.

- **Measured string:** `block3.py` now measures the local parity-odd check $\langle X_0 \rangle$ together with the parity-preserving boundary correlation

$$\langle X_0 Z_1 \cdots Z_{L-2} X_{L-1} \rangle,$$

  which for $L = 4$ is $\langle X_0 Z_1 Z_2 X_3 \rangle$. This matches the diagnostic emphasized in the Week 5 deck and in the later Week 6–7 sweeps.

- **Scope of results:** `block3.py` aims to compare topological, critical, and trivial points ($\mu = 0, 2t, 3t$). The notebook and Week 5 deck only demonstrate the sweet-spot topological case and identify the next task as sweeping μ.

5.4 Comparison with the Existing Notes

The earlier sections of these notes already contain the broad conceptual structure: use VQE for state preparation, avoid treating statevector loading as the final algorithm, and use shot-based sampling for observables. However, the Week 5 material refines several points:

- The statement that the sweet-spot edge observables are simply X_0 and X_{L-1} is too local for the Week 5 demonstration. Single-edge operators flip fermion parity, so their expectation value vanishes in a parity eigenstate. The robust diagnostic is the parity-preserving two-edge correlator.

- The string observable in the shot-based sampling subsection should be aligned with the deck when discussing Week 5:

$$O_{\text{Week 5}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}.$$

  The earlier $Y_{L-1} Z_{L-2} \cdots Z_0$ string corresponds to a different Majorana/string convention and is not the one used in the current presentation.

- The phrase "exact diagonalization must be discarded" should be read carefully. ED should not be the state-preparation algorithm, but both `block3.py` and the notebook still use ED as a small-L validation benchmark for fidelity. That is scientifically reasonable and should remain allowed for validation.

- The Week 5 presentation figure workflow is notebook-first. The PNG figures used by the deck come from `src/block3_week5.ipynb` and are stored under `plots/block3_*.png`, not from the PDF outputs generated by the `block3.py` plot registry.

5.5 Recommended Reconciliation

For the next Block 3 revision, the cleanest path is:

1. Treat `src/block3_week5.ipynb` and `presentation/week5/slides.tex` as the Week 5 snapshot.
2. Refactor `src/block3.py` so it can reproduce the Week 5 sweet spot first: pure YY Hamiltonian, parity-penalized VQE, and measurement of $X_0Z_1 \cdots Z_{L-2}X_{L-1}$.
3. After that, extend the same circuit-measurement protocol to sweep μ and recover the phase diagram from circuit-derived string correlations.
4. Keep exact diagonalization only as a validation path for small L: energy checks, degenerate-subspace fidelity, and debugging of sign/string conventions.

With this reconciliation, the Week 5 deck becomes a valid proof-of-principle: it demonstrates that a VQE-prepared gate-level state can reveal topology through a parity-preserving non-local edge correlator. The broader `block3.py` workflow should then be adjusted to use the same correlator as its primary observable before attempting the full μ sweep.